

Information

Flag question

Assume a binary classification task with classes  $\Omega = \{\text{salmon}, \text{bass}\}$  and three (heavily discretized) features *Length*, *Lightness*, and *Colour*. Also assume we have a training set  $D$  consisting of five fish  $x_i$ :

|       |       |       |      |        |
|-------|-------|-------|------|--------|
| $x_1$ | short | dark  | blue | salmon |
| $x_2$ | short | light | blue | salmon |
| $x_3$ | long  | dark  | red  | salmon |
| $x_4$ | long  | light | red  | bass   |
| $x_5$ | long  | light | red  | bass   |

Time left 0:37:55

Information

Flag question

Consider a  $k$ -NN classifier that uses the simple distance measure " $D(x_i, x_j) = \text{number of feature values in which } x_i \text{ and } x_j \text{ differ}$ ". (Example: in our five fish dataset,  $D(x_3, x_4) = 1$ )

Question 15

Answer saved

Marked out of 2.00

Flag question

What is the accuracy of a 1-NN classifier, when we run a leave-one-out cross-validation on this dataset? (Give the result as a fraction  $a/b$ )

Answer:

Question 16

Answer saved

Marked out of 1.00

Flag question

Will this accuracy estimate change if we add additional copies of  $x_5$  to the dataset?

☐ No

☐ Yes

[Clear my choice](#)

Information

Flag question

Which of the following statements about  $k$ -NN classifiers are correct (**in general**; independently of the above dataset)?

Question 17

Answer saved

Marked out of 1.00

Flag question

When  $k = \text{number of training examples in } D$ , a  $k$ -NN classifier behaves like a baseline classifier

Select one:

☐ True

☐ False

Question 18

Answer saved

Marked out of 1.00

Flag question

For  $k=1$ , a  $k$ -NN classifier can only represent linear decision boundaries

Select one:

☐ True

☐ False

Question 19

Answer saved

Marked out of 1.00

Flag question

Increasing  $k$  reduces the variance of a  $k$ -NN classifier

Select one:

☐ True

☐ False

Which of the following statements about classifier evaluation measures and procedures are true:

Question **10**

Answer saved

Marked out of  
1.00 [Flag question](#)

For all classifiers along the main diagonal of ROC space, recall + precision = 1

Select one:

- ☐ True
- ☐ False

Question **11**

Answer saved

Marked out of  
1.00 [Flag question](#)

Two classifiers with the same classification error on a particular test set may differ in recall and precision

Select one:

- ☐ True
- ☐ False

Question **12**

Answer saved

Marked out of  
1.00 [Flag question](#)

All classifiers along the main diagonal of ROC space have the same error

Select one:

- ☐ True
- ☐ False

Question **13**

Answer saved

Marked out of  
1.00 [Flag question](#)

When we evaluate a perfectly random classifier (that chooses its class prediction by sampling from a uniform distribution) via cross-validation on some dataset, we may get an estimate that is higher than the baseline.

Select one:

- ☐ True
- ☐ False

Time left 0:38:09

Question **14**

Answer saved

Marked out of  
1.00 [Flag question](#)

Two classifiers with the same recall and the same precision on a particular test set may differ in classification accuracy

Select one:

- ☐ True
- ☐ False

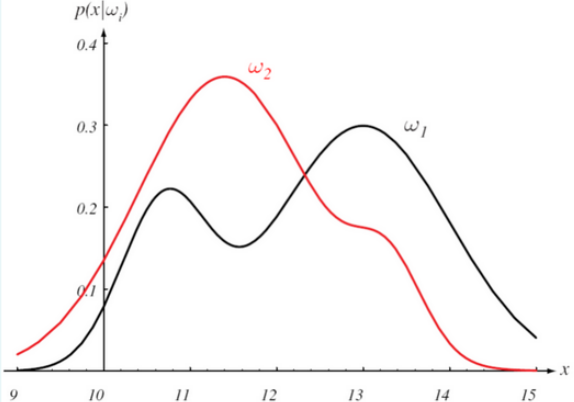


Figure 1: Class-conditional probability density functions  $p(x \mid \omega_i)$  show the probability density of measuring a particular value  $x$  given that the pattern belongs to category  $\omega_i$ . If  $x$  is the lightness of the fish, the two curves might describe the difference in lightness of populations of two types of fish. Density functions are normalised, and thus the area under each curve is 1.0. (From [Duda *et al.*, 2001].)

Consider the class-conditional distributions over a single feature  $x$  from our lecture slides, replicated above (along with the figure caption from the slides), and consider the task of learning a classifier that can predict classes  $\omega_1$  and  $\omega_2$  from this feature.

Which of the following statements are true in a world described by these distributions:

In this case,  $\omega_1$  and  $\omega_2$  are equally probable (because the area under both curves is 1.0)

Select one:

- ☐ True
- ☐ False

If the prior class distribution were uniform ( $P(\omega_1) = P(\omega_2) = 1/2$ ), the two curves would be perfectly flat (constant)

Select one:

- ☐ True
- ☐ False

If we change the prior distribution between the classes, these two curves will also change

Select one:

- ☐ True
- ☐ False

Given these curves,  $\omega_1$  is slightly more probable than  $\omega_2$

Select one:

- ☐ True
- ☐ False

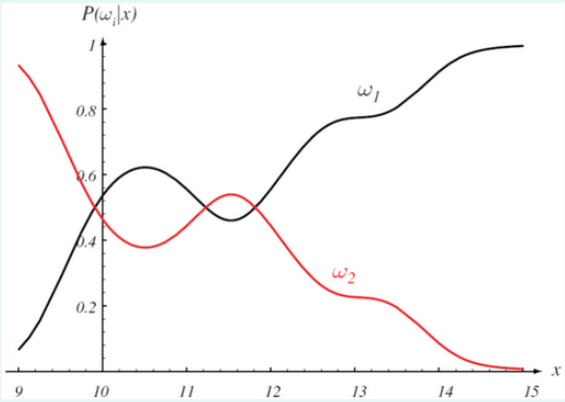


Figure 2: Posterior probabilities (**for the particular priors**  $P(\omega_1) = 2/3$  and  $P(\omega_2) = 1/3$ ) following from the class-conditional probability densities shown in the previous figure. If an object is measured to have, e.g., feature value  $x = 14$ , the probability that it is in category  $\omega_2$  is roughly 0.08, and that it is in  $\omega_1$  is 0.92. At every  $x$ , the posteriors sum to 1.0. (From [Duda *et al.*, 2001].)

Now consider MAP or Minimum Risk Classification from the same single feature  $x$ , using the same class-conditional distributions described before (Figure 1), which gives the following posterior distributions (again taken from the lecture slides, see Figure 2). Note that this is assuming a certain prior probability distribution over the classes ( $P(\omega_1) = 2/3$ ,  $P(\omega_2) = 1/3$ ).

Which of the following statements are true?

If the prior class distribution were uniform ( $P(\omega_1) = P(\omega_2) = 1/2$ ), the two curves would become perfectly flat (constant)

Select one:

- ☐ True
- ☐ False

There is no prior distribution under which these two posterior curves (based on the class-conditional distributions in Figure 1) could become identical.

Select one:

- ☐ True
- ☐ False

If the red curve were to reach exactly zero, the black curve would go to infinity.

Select one:

- ☐ True
- ☐ False

Time left 0:38:30

If we changed the prior so that  $P(\omega_2) > 0.5$ , the two posterior curves would no longer sum to 1.0 at every  $x$

Select one:

- ☐ True
- ☐ False

If the cost of misclassifying examples of  $\omega_1$  as  $\omega_2$  were 3 times as high as the cost of misclassifying  $\omega_2$  as  $\omega_1$ , the black line in the figure would be raised (and the red one lowered).

Select one:

- ☐ True
- ☐ False

Which of the following statements about Markov and Hidden Markov Models are true?

A Markov Model can model state sequences of arbitrary length.

Select one:

- ☐ True
- ☐ False

A sequence of coin tosses (possible outcomes: heads/tails, one coin, one person) satisfies the Markov property.

Select one:

- ☐ True
- ☐ False

In a Hidden Markov Model, there cannot be more states than different observation symbols.

Select one:

- ☐ True
- ☐ False

Without the Time Invariance property, the state transition matrix  $A$  of a Markov model could not be written as an  $S \times S$  matrix.

Select one:

- ☐ True
- ☐ False

What is the likelihood of a HMM  $M$  relative to a sequence of observations  $O$  ?

- ☐  $P(O \mid M)$
- ☐  $\arg \max_{S_i} P(S_i \mid \langle o_1, \dots, o_t \rangle)$
- ☐  $\arg \max_{S_i} P(S_i \mid O)$
- ☐  $P(M \mid O)$
- ☐  $\arg \max_i P(M_i \mid O)$
- ☐  $\arg \max_{\theta} P(O \mid M_{\theta})$

Consider the task of identifying the current season (winter, spring, summer, autumn), given a sequence of daily humidity measurements. Assume we have learned 4 Hidden Markov Models (HMMs)  $M_i, i = 1..4$ , one for each season, that model the weather in each season in terms of 3 hidden weather states (rainy, sunny, cloudy) and 3 observable humidity levels (dry, medium, humid).

Which of the following quantities needed for classifying the season of a specific observation sequence  $O$  by Bayesian MAP classification must be estimated from each model  $M_i$ ?

- ☐  $P(M_i)$
- ☐  $\arg \max_{\theta} P(O \mid M_{i,\theta})$
- ☐  $P(O \mid M_i)$
- ☐  $P(M_i \mid O)$

If one season is longer than the others, and we assume that the number of observation sequences for a particular season that the model will encounter is proportional to the season length, how would you take this into account in MAP classification via Bayes' rule?

- ☐ by weighting the training examples
- ☐ via a cost function
- ☐ via different prior probabilities
- ☐ by undersampling the longer season

Question **35**

Answer saved

Marked out of 1.00

🚩 [Flag question](#)

Consider a world (feature space) defined by 100 numeric (real-valued) features and a dataset  $D$  with 1,000 examples (feature vectors). How many numbers (parameters) have to be estimated when fitting a 5-component Gaussian Mixture Model (GMM) onto this dataset? (ignoring possible symmetries, which might make some numbers redundant ..)

Answer:

Question **36**

Answer saved

Marked out of 1.00

🚩 [Flag question](#)

How does the number of parameters that need to be estimated for a GMM grow with the number of features?

- ☐ exponentially
- ☐ linearly
- ☐ logarithmically
- ☐ quadratically
- ☐ not at all; it would only grow with growing  $k$

Question **37**

Answer saved

Marked out of 1.00

🚩 [Flag question](#)

How does the number of parameters that need to be estimated for a GMM grow with the number of examples?

- ☐ quadratically
- ☐ logarithmically
- ☐ not at all
- ☐ exponentially
- ☐ linearly

Time left 0:37:25

Information

🚩 [Flag question](#)

What do k-means clustering and GMM fitting have in common?

Question **38**

Answer saved

Marked out of 1.00

🚩 [Flag question](#)

They can both be used to learn the likelihood term needed for classification via Bayes' rule.

Select one:

- ☐ True
- ☐ False

Question **39**

Answer saved

Marked out of 1.00

🚩 [Flag question](#)

They are both maximum likelihood methods.

Select one:

- ☐ True
- ☐ False

Question **40**

Answer saved

Marked out of 1.00

🚩 [Flag question](#)

They are both learned by an EM-style model fitting algorithm.

Select one:

- ☐ True
- ☐ False

Question **41**

Answer saved

Marked out of 1.00

🚩 [Flag question](#)

They both work on unlabeled points from feature space.

Select one:

- ☐ True
- ☐ False

Question **42**

Answer saved

Marked out of 1.00

🚩 [Flag question](#)

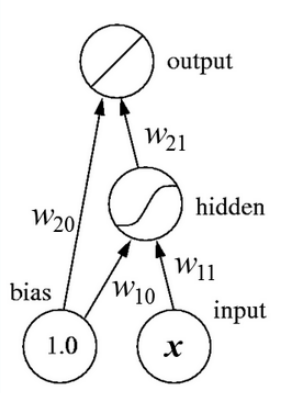
They both partition the data into  $k$  subsets.

Select one:

- ☐ True
- ☐ False

Consider the simple neural network from the lecture slides, with 1 input unit (for a feature  $x$ ), one hidden unit, and one output unit  $y$ , where the hidden unit uses the tanh as activation function, and the output unit  $y$  uses the identity function (i.e., just outputs what comes in as net input). Also, assume that the loss function is the sum squared error (as defined in the slides).

Which of the following statements are correct?



The net input to the hidden unit is a linear function of  $x$

Select one:

- ☐ True
- ☐ False

When the output  $\hat{y}$  calculated by the network for a given training input  $x$  is correct (i.e., equal to the true  $y$  value of the training example), stochastic gradient descent will not change any of the weights.

Select one:

- ☐ True
- ☐ False

During error back-propagation, the gradient wrt.  $w_{20}$  will usually be computed before the gradient wrt.  $w_{10}$ .

Select one:

- ☐ True
- ☐ False

Changing the tanh activation function in the hidden unit to make it steeper has the same effect as increasing the learning rate

Select one:

- ☐ True
- ☐ False

Now let's make this network 102 layers deep, with one input layer (unit)  $x$ , one output layer (unit)  $y$ , and 100 hidden layers, each consisting of one unit with tanh activation function.

This deep network has ... parameters (weights):

Answer:

In backpropagation learning, the gradients w.r.t. the weights in the lowest level of this deep network will be only 1/100 of corresponding gradients in the simple 3-level model (vanishing gradient problem)

Select one:

- ☐ True
- ☐ False

Time left 0:37:42

This deep network can approximate any mathematical function of one argument ( $x$ ) to any degree of accuracy (if tanh is a universal basis function)

Select one:

- ☐ True
- ☐ False

This network can learn the same set of functions as a network with 1 hidden layer consisting of 100 units.

Select one:

- ☐ True
- ☐ False

Assume a binary classification task with classes  $\Omega = \{\text{salmon}, \text{bass}\}$  and three (heavily discretized) features *Length*, *Lightness*, and *Colour*.

Also assume we have a training set D consisting of five fish  $x_i$ :

|       |       |       |      |        |
|-------|-------|-------|------|--------|
| $x_1$ | short | dark  | blue | salmon |
| $x_2$ | short | light | blue | salmon |
| $x_3$ | long  | dark  | red  | salmon |
| $x_4$ | long  | light | red  | bass   |
| $x_5$ | long  | light | red  | bass   |

Answer the following questions **in the context of this specific dataset D**:

If we duplicate the dataset D by adding a copy of each training example, the resulting decision tree learned by ID3 will remain the same

Select one:

- ☐ True
- ☐ False

There is no decision tree that is consistent with these 5 examples

Select one:

- ☐ True
- ☐ False

What is the depth of the decision tree that ID3 will learn from D? (The depth of the root is zero)

Answer:

Answer the following questions **in general** (independent of the above dataset):

Consider a variant "ID3-R" of ID3 that uses a random number generator instead of the information gain formula for feature selection. Is the following statement about ID3-R true or false:

Trees learned by ID3-R cannot overfit the training data

Select one:

- ☐ True
- ☐ False

Consider the following modification to the ID3 algorithm in the lecture slides: "Turn a node into a leaf if the entropy of the class distribution in that node is  $\leq 0.1$ ."

Will this increase the bias of the algorithm?

- ☐ a. yes
- ☐ b. no

Duplicating training examples (adding additional copies of some examples to the training set) does not change the tree ID3 will learn

Select one:

- ☐ True
- ☐ False

Consider a variant "ID3-R" of ID3 that uses a random number generator instead of the information gain formula for feature selection. Is the following statement about ID3-R true or false:

ID3-R is not a valid learning algorithm

Select one:

- ☐ True
- ☐ False